

Thermal conductivity of graphene-reinforced polymer composites: a surrogate modelling approach

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Abstract

Predicting the thermal conductivity of polymer composites reinforced with graphene platelets is important for the design of thermal management materials. In this work, we develop a gradient boosting regressor (GBR) surrogate model trained on a synthetic dataset of 120 specimens with eight input variables. The model achieves a coefficient of determination of $R^2 = 0.847 \pm 0.031$ on the test set, outperforming a linear regression baseline ($R^2 = 0.631 \pm 0.044$). Feature importance analysis identifies filler content as the dominant predictor (38.4% importance), followed by aspect ratio (27.1%). These results suggest that surrogate modelling is a viable approach for rapid property screening in graphene composite design.

DISCLAIMER: All data, values, and references in this manuscript are synthetic. This is an illustrative example for the scientific-redaction-skills framework.

Keywords: thermal conductivity, polymer composite, graphene, surrogate model, gradient boosting, machine learning

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1. Introduction

Graphene-reinforced polymer composites have attracted considerable interest as thermal management materials over the past two decades, driven by graphene’s exceptional in-plane thermal conductivity of approximately $5000 \text{ W m}^{-1} \text{ K}^{-1}$ [?]. In contrast, the polymer matrices used in these composites typically exhibit conductivities in the range $0.1\text{--}0.3 \text{ W m}^{-1} \text{ K}^{-1}$ [?], creating a strong thermodynamic incentive to engineer the filler–matrix interface for maximum heat transfer.

Despite two decades of experimental study, the relationship between processing variables and composite thermal conductivity remains difficult to predict from first principles alone. Effective medium theories [?] provide analytical bounds but do not capture the complex dependence on filler content, aspect ratio, interface resistance, and matrix morphology simultaneously [?]. As a result, property optimisation has traditionally required extensive experimental campaigns, which are both time-consuming and resource-intensive.

Surrogate modelling approaches—also known as metamodeling—offer an alternative path for accelerating materials design [?]. By training a data-driven model on a curated dataset of experimental or simulation results, a surrogate can predict material properties at a fraction of the computational cost of physics-based models [?]. Applications of surrogate modelling to composite thermal conductivity have been reported for carbon-nanotube-reinforced matrices [?], but studies specifically targeting graphene platelet composites remain limited.

In this work, we develop a gradient boosting regressor (GBR) surrogate model trained on a dataset of 120 synthetic specimens characterised by eight

input variables. The objectives of this study are: (i) to evaluate the predictive accuracy of the GBR model on held-out test data; (ii) to identify the most influential input variables using feature importance analysis; and (iii) to characterise the prediction uncertainty across the experimental design space.

2. Materials and Methods

2.1. Dataset

The dataset comprised 120 specimens described by eight input variables: filler content (vol%), aspect ratio, matrix thermal conductivity ($\text{W m}^{-1} \text{K}^{-1}$), interface thermal resistance ($\text{m}^2 \text{K W}^{-1}$), processing temperature ($^{\circ}\text{C}$), mixing time (min), filler surface treatment (binary), and specimen thickness (mm). The output variable was measured in-plane thermal conductivity ($\text{W m}^{-1} \text{K}^{-1}$).

All values in the dataset are synthetic and were generated to illustrate the surrogate modelling workflow; they have no physical validity.

2.2. Train/test split

The dataset was divided into a training set of 96 specimens (80%) and a test set of 24 specimens (20%) using a random split with a fixed random seed. No stratification was applied.

2.3. Surrogate model

A gradient boosting regressor (GBR) was implemented using an illustrative software package. The key hyperparameters were set as follows: number

of estimators = 200, maximum tree depth = 5, learning rate = 0.05, minimum samples per leaf = 4. These values were selected via a preliminary grid search on the training set (details in Supplementary Section S1).

2.4. Cross-validation and evaluation

Model performance was evaluated using 5-fold cross-validation on the training set. The final model was then evaluated on the held-out test set. Performance metrics reported are the coefficient of determination (R^2), root mean square error (RMSE, $\text{W m}^{-1} \text{K}^{-1}$), and mean absolute error (MAE, $\text{W m}^{-1} \text{K}^{-1}$). Uncertainty in reported metrics was estimated as the standard deviation across the five cross-validation folds.

2.5. Feature importance analysis

Feature importance was estimated using mean decrease in impurity (MDI) averaged over all 200 trees in the GBR ensemble.

2.6. Software and reproducibility

All analyses were performed in Python (version X.Y, ILLUSTRATIVE). The dataset and model configuration files are provided as supplementary material (Supplementary Data S1 and S2, ILLUSTRATIVE).

3. Results

3.1. Model accuracy on the test set

The GBR model achieved $R^2 = 0.847 \pm 0.031$ and $\text{RMSE} = 0.143 \pm 0.018 \text{ W m}^{-1} \text{K}^{-1}$ on the held-out test set (Table ??). The mean absolute error was $0.112 \pm 0.014 \text{ W m}^{-1} \text{K}^{-1}$. The training set R^2 was 0.924 ± 0.012 ,

indicating moderate overfitting that remained within acceptable limits for a 120-sample dataset.

3.2. Comparison to the linear regression baseline

The GBR model outperformed the linear regression baseline across all metrics (Table ??). The baseline achieved $R^2 = 0.631 \pm 0.044$ on the test set, compared to 0.847 ± 0.031 for GBR, corresponding to a difference of 0.216 in R^2 .

Table 1: Performance metrics for the GBR and linear regression baseline models on the held-out test set (120 specimens, 80/20 split, 5-fold CV). ILLUSTRATIVE.

Metric	GBR	Linear regression
R^2	0.847 ± 0.031	0.631 ± 0.044
RMSE ($\text{W m}^{-1} \text{K}^{-1}$)	0.143 ± 0.018	0.241 ± 0.031
MAE ($\text{W m}^{-1} \text{K}^{-1}$)	0.112 ± 0.014	0.198 ± 0.025

3.3. Feature importance

Filler content was identified as the dominant predictor, with a mean decrease in impurity of 38.4% across all 200 trees (Figure ??). Aspect ratio ranked second at 27.1%, followed by matrix thermal conductivity at 19.6%. The remaining five variables collectively accounted for 14.9% of total importance.

3.4. Sensitivity analysis

The model prediction range across all specimens spanned 0.21–3.84 $\text{W m}^{-1} \text{K}^{-1}$, consistent with the physical range reported for graphene-filled polymer composites in the synthetic design space (Supplementary Table S1).

[Figure 1 placeholder — feature importance bar chart — ILLUSTRATIVE]

Figure 1: Feature importance for the GBR model estimated by mean decrease in impurity over 200 trees. Filler content dominates with 38.4% importance, followed by aspect ratio (27.1%) and matrix conductivity (19.6%). Dataset: 120 synthetic specimens. ILLUSTRATIVE.

4. Discussion

The test set R^2 of 0.847 ± 0.031 suggests that the GBR surrogate generalises reasonably well to compositions not seen during training, given the limited dataset size of 120 specimens. The gap between training (0.924) and test (0.847) performance is consistent with mild overfitting, which may be reduced by increasing the dataset size or applying additional regularisation.

The dominance of filler content in the feature importance analysis is consistent with effective medium theory predictions, which assign primary importance to the volume fraction of the filler phase [?]. The secondary role of aspect ratio aligns with percolation-threshold arguments for high-aspect-ratio fillers [?].

The observed performance degradation at high filler loadings (Supplementary Section S2) may be attributed to filler agglomeration, which reduces the effective contact area between graphene platelets and the matrix. This interpretation is speculative and should be investigated with targeted experiments before being stated as a conclusion.

Limitations.. This study is limited to the synthetic design space covered by the 120-specimen dataset. The model cannot be expected to extrapolate reliably beyond the training distribution. The feature importance estimates are

specific to the GBR architecture and may not reflect the physical importance of each variable in a broader context. All data in this study are synthetic; the results have no direct applicability to real graphene composite systems.

5. Conclusions

A gradient boosting regressor surrogate model was developed for predicting the thermal conductivity of graphene-reinforced polymer composites in a synthetic dataset. The model achieved $R^2 = 0.847 \pm 0.031$ on the test set, outperforming a linear regression baseline ($R^2 = 0.631 \pm 0.044$). Filler content was identified as the dominant predictor (38.4% feature importance). These results demonstrate the feasibility of surrogate modelling as a screening tool for composite property optimisation, within the limitations stated above.

AI use disclosure.. This illustrative manuscript was prepared with the assistance of the scientific-redaction-skills framework (Claude Code, Anthropic). All scientific decisions, claims, and content are the responsibility of the authors. *(This is an illustrative disclosure statement for the example project.)*

Supplementary Material

Table of Contents

- Section S1: Hyperparameter optimisation details
- Section S2: Sensitivity analysis at high filler loadings
- Table S1: Full synthetic dataset statistics
- Figure S1: Predicted vs. observed parity plot

S1. Hyperparameter optimisation

A grid search was performed over the following hyperparameter space on the training set using 5-fold cross-validation: number of estimators $\in \{100, 200, 300\}$, maximum depth $\in \{3, 5, 7\}$, learning rate $\in \{0.01, 0.05, 0.1\}$. The optimal configuration ($n_{\text{est}} = 200$, depth = 5, lr = 0.05) was selected based on the lowest mean RMSE across folds.

S2. Sensitivity analysis at high filler loadings

Model predictions at filler contents above 15 vol% showed increased variance relative to the low-loading regime (Figure S1). This pattern is consistent with the physical hypothesis of agglomeration effects at high loadings, as discussed in Section 4.

[Figure S1 placeholder — parity plot, predicted vs. observed, ILLUSTRATIVE]

Figure S1: Predicted vs. observed thermal conductivity for the GBR model on the test set (24 specimens). Dashed line: perfect agreement. Points at high predicted conductivity ($> 2.5 \text{ W m}^{-1} \text{ K}^{-1}$) show increased scatter, corresponding to high-filler-loading specimens. All values are synthetic. ILLUSTRATIVE.

Table S1: Summary statistics of the synthetic dataset (120 specimens). ILLUSTRATIVE.

Variable	Min	Mean	Max
Filler content (vol%)	0.5	8.3	20.0
Aspect ratio	10	142	500
Matrix conductivity ($\text{W m}^{-1} \text{ K}^{-1}$)	0.10	0.21	0.32
Output: thermal conductivity ($\text{W m}^{-1} \text{ K}^{-1}$)	0.21	1.42	3.84